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CLASS: CS2

practical 3
3.1.1
Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using ndim
- The shape of the array using shape
- The total number of elements in the array using size

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, and , representing the number of rows and the number of columns.
- The next lines contain space-separated integers representing the elements of the array, entered row by row.

Output Format:

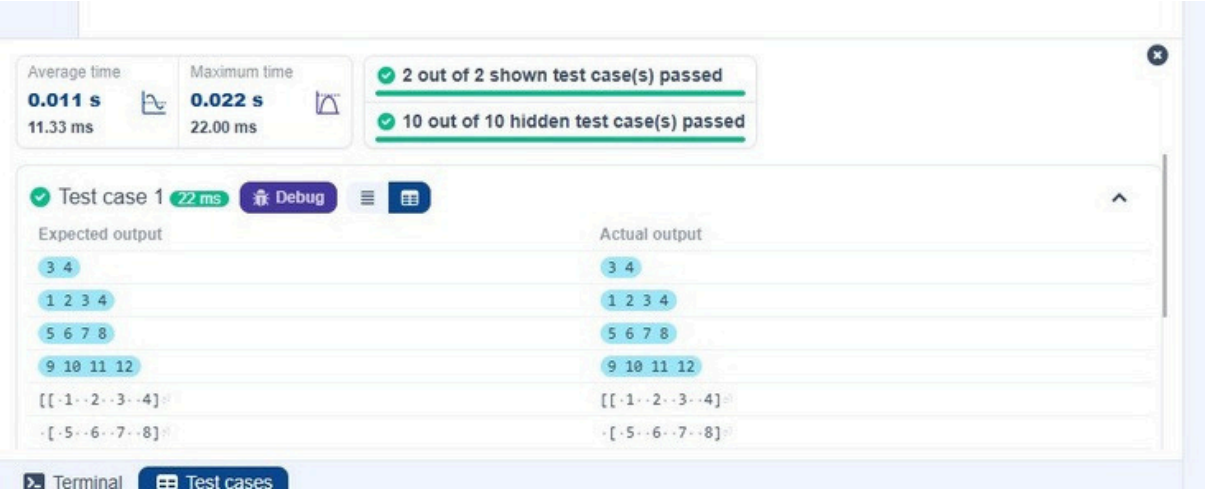
- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use reshape() function to reshape the input array with the specified number of rows and columns.

```
import numpy as np

rows,cols = map(int,input().split())
```

```
elements = []
for _ in range(rows) :
        elements.extend(map(int,input().split()))

array = np.array(elements).reshape(rows,cols)
print(array)
print(array.ndim)
print(array.shape)
print(array.size)
```

output :



3.2.1

The given code takes two matrices, , and , as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. **Addition** ()
2. **Subtraction** ()
3. **Element-wise Multiplication** ()
4. **Matrix Multiplication** ()

5. **Transpose of Matrix A**

Input Format:

- The user will input 3 rows for , each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for , each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

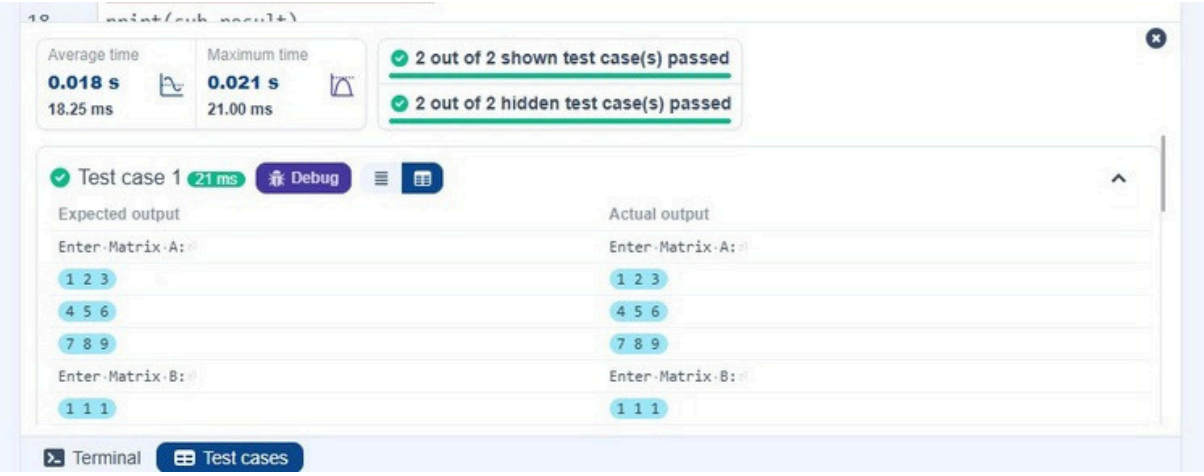
1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

code :

```
import numpy as np

#Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Matrix B:")
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])

#Addition
addition_result = matrix_a +matrix_b
```

```
print("Addition (A + B):")
print(addition_result)
# Subtraction
sub_result = matrix_a - matrix_b
print("Subtraction (A - B):")
print(sub_result)
# Multiplication (element-wise)
elemental_multi_result = matrix_a * matrix_b
print("Element-wise Multiplication (A * B):")
print(elemental_multi_result)
# Matrix multiplication (dot product)
matrix_dotproduct_result = np.dot(matrix_a , matrix_b )
print("A dot B:")
print(matrix_dotproduct_result)
# Transpose
transpose_a = matrix_a.T
print("Transpose of A:")
print(transpose_a)
```

output :



3.2.2

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking**: Stack the two matrices horizontally (side by side).
- **Vertical Stacking**: Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Code :

```
import numpy as np


#Input matrices
```

```
print("Enter Array1:")
arr1 = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Array2:")
arr2 = np.array([list(map(int, input().split())) for i in range(3)])

#Perform horizontal stacking (hstack)
h_stack = np.hstack((arr1, arr2))
print("Horizontal Stack:")
print(h_stack)
```

```
#Perform vertical stacking (vstack)
v_stack = np.vstack((arr1, arr2))
print("Vertical Stack:")
print(v_stack)
```

output :

3.2.3

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Code:

```
import numpy as np


#Take user input for the start, stop, and step of the sequence

start = int(input())

stop = int(input())

step = int(input())


#Generate the sequence using np.arange()

sequence = np.arange(start, stop, step)

# Print the generated sequence

print(sequence)
```

output:

3.2.4

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Code :

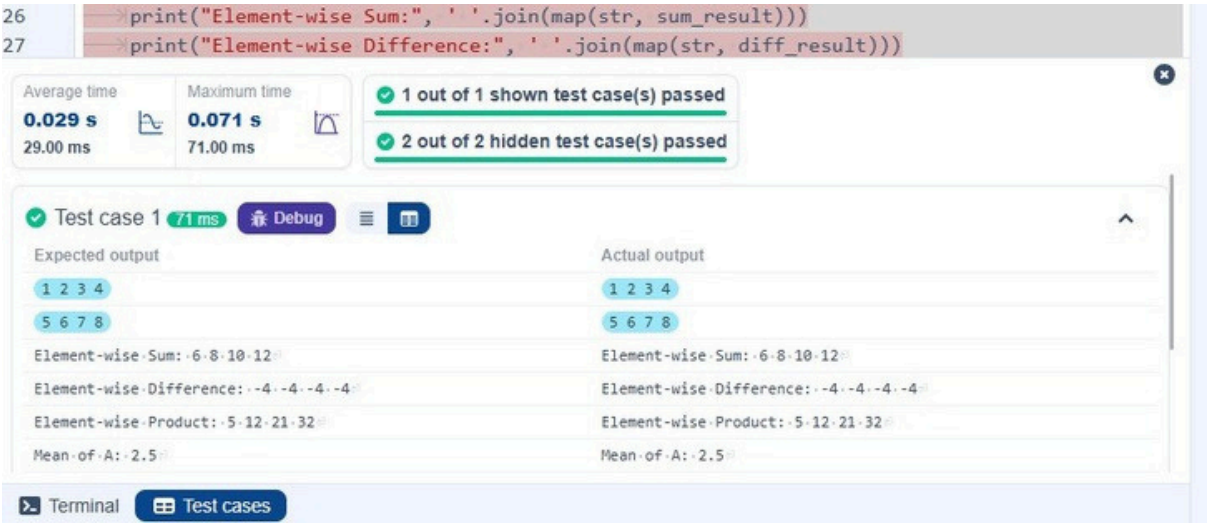
```
import numpy as np
```

```
defarray_operations(A, B):

        #Convert A and B to NumPy arrays
        A= np.array(A)
        B= np.array(B)



        # Arithmetic Operations
        sum_result = A + B
        diff_result = A - B
        prod_result = A * B

        # Statistical Operations
        mean_A = np.mean(A)
        median_A =np.median(A)
        std_dev_A = np.std(A)

        # Bitwise Operations
        and_result = A & B
        or_result = A |B
        xor_result = A ^ B

   #Output results with one space between each element
        print("Element-wise Sum:", ' '.join(map(str, sum_result)))
        print("Element-wise Difference:", ' '.join(map(str, diff_result)))
        print("Element-wise Product:", ' '.join(map(str, prod_result)))
```

```
print(f"Mean of A: {mean_A}")
print(f"Median of A: {median_A}")
print(f"Standard Deviation of A: {std_dev_A}")

print("Bitwise AND:", ' '.join(map(str, and_result)))
print("Bitwise OR:", ' '.join(map(str, or_result)))
print("Bitwise XOR:", ' '.join(map(str, xor_result)))
```

```
A=list(map(int, input().split())) # Elements of array A
B=list(map(int, input().split())) # Elements of array B
array_operations(A, B)
```

output:



3.2.5

he given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <**original_array**>

View array: <**view_array**>

- After modifying the copy:

Original array after modifying copy: <**original_array**>

Copy array: <**copy_array**>

Code and output :

```python
import numpy as np

inputlist = list(map(int,input().split(" ")))

# Original array
original_array = np.array(inputlist)

# Create a view
view_array = original_array.view()

# Create a copy
copy_array = original_array.copy()

# Modify the view
view_array[0] = 99
print("Original array after modifying view:", original_array)
print("View array:", view_array)
```

Average time 0.009 s 9.25 ms | Maximum time 0.016 s 16.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 16 ms

Test case 2 6 ms

3.2.6

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- search_value: The value to search for in the array.

- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

1. Searching: Find the indices where search_value appears in array1 and print these indices.

2. Counting: Count how many times count_value appears in array1 and print the count.

3. Broadcasting: Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.

4. Sorting: Sort array1 in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing array1.
2. An integer search_value represents the value to search for in the array.
3. An integer count_value represents the value to count in the array.
4. An integer broadcast_value represents the value to add to each element of the array.

Output Format:

1. The indices where search_value occurs in array1.
2. The count of occurrences of count_value in array1.
3. The array after adding the broadcast_value to each element.
4. The sorted array.

code and output :

```
import numpy as np

# Input array from the user
array1 = np.array(list(map(int, input().split())))

# Searching
search_value = int(input("Value to search: "))
count_value = int(input("Value to count: "))
broadcast_value = int(input("Value to add: "))

# Find indices where value matches in array1
a=np.where(array1==search_value)[0]
print(a)
# Count occurrences in array1
b=np.count_nonzero(array1==count_value)
print(b)

# Broadcasting addition
c=array1 + broadcast_value
print(c)

# Sort the first array
d=np.sort(array1)
print(d)
```

```
Average time
0.026 s
26.00 ms
Maximum time
0.057 s
57.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
Debug
Expected output
1 1 1 2 2 2
Value to search: 1
Value to count: 2
Value to add: 2
[0 1 2]
3
Actual output
1 1 1 2 2 2
Value to search: 1
Value to count: 2
Value to add: 2
[0 1 2]
3
Terminal
Test cases
```

3.2.7

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details**: Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students**: Determine the total number of students in the dataset.
- **Print all student roll numbers**: Extract and print the roll numbers of all students.
- **Print Subject 1 marks**: Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2**: Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3**: Identify the highest marks in Subject 3.
- **Print all subject marks**: Display the marks of all students for each subject.
- **Find total marks of students**: Compute the total marks for each student across all subjects.
- **Find the average marks of each student**: Compute the average marks for each student.
- **Find average marks of each subject**: Compute the average marks for all students in each subject.

- **Find average marks of Subject 1 and Subject 2**: Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3**: Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3**: Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2**: Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2**: Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1**: Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2**: Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored >=90 in each subject**: Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored >=90**: Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order**: Sort and print the marks of students in Subject 1 in ascending order.
- **Print students who scored between 50 and 90 in Subject 1**: Display students who scored marks between 50 and 90 in Subject 1.
- **Find index positions of students who scored 79 in Subject 1**: Identify the index positions of students who scored exactly 79 marks in Subject 1.
-

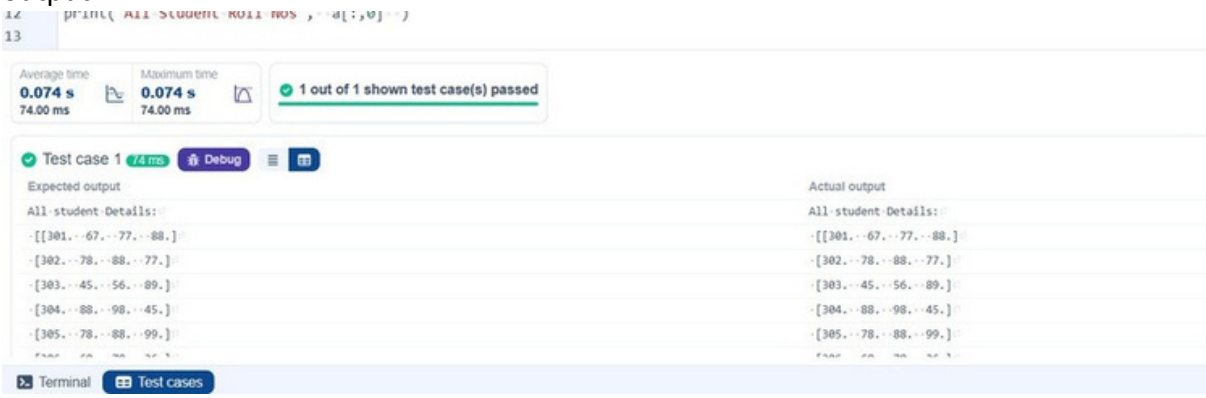
Note: Fill in the missing code to perform the above-mentioned operations.

Code :

```
import numpy as np


a= np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

#1. Print all student details

print("All student Details:\n", a )
```

```
#2. print total students

print("Total Students:",a.shape[0] )

#3. Print all student Roll numbers
print("All Student Roll Nos", a[:,0] )

#4. Print subject 1 marks
print("Subject 1 Marks", a[:,1])

#5. print minimum marks of Subject 2
print("Min marks in Subject 2", np.min(a[:,2] ))

#6. print maximum marks of Subject 3
print("Max marks in Subject 3", np.max(a[:,3])    )

#7. Print All subject marks
print("All subject marks:", a[:,1:]     )

#8. print Total marks of students
print("Total Marks", np.sum(a[:,1:],axis=1)       )

#9. print average marks of each student
avg=np.mean(a[:,1:],axis=1)
print(np.round(avg,1))

#10. print average marks of each subject
```

```
print("Average Marks of each subject", np.mean(a[:,1:],axis=0)      )

#11. print average marks of S1 and S2
print("Average Marks of S1 and S2", np.mean(a[:,1:3],axis=0 )      )

#12. print average marks of S1 and S3
print("Average Marks of S1 and S3", np.mean(a[:,[1,3]],axis=0)             )

#13. print Roll number who got maximum marks in Subject 3
i=np.argmax(a[:,3])
print("Roll no who got maximum marks in Subject 3", a[i,0]         )

#14. print Roll number who got minimum marks in Subject 2
j=np.argmin(a[:,2])
print("Roll no who got minimum marks in Subject 2", a[j,0] )

#15. print Roll number who got 24 marks in Subject 2

print(f"Roll no who got 24 marks in Subject 2 [{a[a[:,2]==24,0]}]"       )

#16. print count of students who got marks in Subject 1 < 40
print("Count of students who got marks in Subject 1 < 40", np.sum(a[:,1]<40)            )

#17. print count of students who got marks in Subject 2 > 90
print("Count of students who got marks in Subject 2 > 90:", np.sum(a[:,2]>90)            )
Count=[len(np.argwhere(a[:,1]>=90)),len(np.argwhere(a[:,2]>=90)),len(np.argwhere(a[:,3]>=
90))]
count=np.array(Count)
```

```
# 18. print count of students in each subject who got marks >= 90
print("Count of students in each subject who got marks >= 90:", count          )

#19. print count of subjects in which each student got marks >= 90
print("Roll no:", a[:,0]     )
print("Count of subjects in which student got marks >= 90:", np.sum(a[:,1:]>=90,axis=1)      )

#20. Print S1 marks in ascending order
print(np.sort(a[:,1]))
print(a[(a[:,1]>=50)     &     (a[:,1]<=90)])
print(a[(a[:,1]>=45)     &     (a[:,1]<=90)])
k=np.where(a[:,1]==79) print(k)
```

output :